



KA-EDM

Electroconductive Alumina Composite

KEY PROPERTIES

- Very high hardness
- Excellent strength / toughness balance
- High thermal conductivity
- Ra < 0.1 µm after mirror polishing
- Machinable by EDM (electro-discharge machining)

APPLICATIONS

- Cutting tools
- Wear resistant tools for stamping
- Cold drawing dies
- Machinery parts
- Ballistic protection
- Chemical industry

MAIN PROPERTIES

Property	Value	Unit
Density	4.2	g/cm ³
Flexural Strength	850	MPa
Hardness	22.5	GPa
Fracture Toughness	8.0	MPa·Vm
Young Modulus	370	GPa
Thermal Conductivity (100°C)	35	W/m·K
CTE (20–1000°C)	8.0	x10 ⁻⁶ K ⁻¹

* All properties measured at 20°C unless otherwise stated.

** Measured in a wall thickness of 2 mm

OPERATING CONDITIONS

Max working temp (air)	1000 °C
Recomended temp	<950 °C
Continuous temp	900 °C

ELECTRICAL BEHAVIOR

Resistivity	3.1·10 ⁻⁵ Ω·cm
Conductivity	High

MATERIAL INFORMATION

Material type	Electroconductive ceramic
Machinability	EDM machinable
Structure	Composite ceramic
Key facture	Conductive

Local overheating may occur at electrical contact points due to current concentration and geometry effects.

The information provided is for guidance only and does not constitute a guarantee.

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